



AQ3.1: Activity Questions 1 - Not Graded



This assignment will not be graded and is only for practice.

Level 1

1 point

Consider the following expression:

$v_1 = (1, 0), v_2 = (0, 0), v_3 = (0, 1), v_5 = 1, v_6 = 5, v_7 = (1, 0, 1), v_8 = (0, 0, 1), v_9 = (0, 0, 0), v_{10} = (0, 1, 0), v_1 = (1, 0), v_2 = (0, 0), v_3 = (0, 1), v_5 = 1, v_6 = 5, v_7 = (1, 0, 1), v_8 = (0, 0, 1), v_9 = (0, 0, 0), v_{10} = (0, 1, 0)$. Choose the set of correct options.

☐

v_1, v_7 represent vectors in R^2 v_1, v_7 represent vectors in R^2 .

☐

v_2, v_5 represent vectors in R^1 v_2, v_5 represent vectors in R^1 .

☐

v_2, v_5, v_9 represent vectors in R^3 v_2, v_5, v_9 represent vectors in R^3 .

☐

v_1, v_2, v_3 represent vectors in R^2 v_1, v_2, v_3 represent vectors in R^2 .

☐

v_5, v_6 represent vectors in R^1 v_5, v_6 represent vectors in R^1 .

☐

v_1, v_6 represent vectors in R^1 v_1, v_6 represent vectors in R^1 .

☐

v_8, v_9, v_{10} represent vectors in R^3 v_8, v_9, v_{10} represent vectors in R^3 .

No, the answer is incorrect.

Score: 0

Accepted Answers:

v_1, v_2, v_3 represent vectors in R^2 v_1, v_2, v_3 represent vectors in R^2 .

v_5, v_6 represent vectors in R^1 v_5, v_6 represent vectors in R^1 .

v_8, v_9, v_{10} represent vectors in R^3 v_8, v_9, v_{10} represent vectors in R^3 .

1 point

Let v, v_1, v_2, v, v_1, v_2 represent vectors in a vector space V over R . Which of the following expressions make sense and represent vectors in V ?

☐

$cv \in V$ $cv \in V$, where $c \in R$ $c \in R$.

☐

$\frac{v}{2}v$.

☐

$v^2 v_2$.

☐

$v_1 - v_2 v_1 - v_2$.

☐

$v_1^2 - v_2^2 v_1^2 - v_2^2$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$cv \in V$ $cv \in V$, where $c \in R$ $c \in R$.

$\frac{v}{2}v$.



$$v_1 - v_2 \quad v_1 - v_2.$$

1 point

Consider the set $V = \{(z - x, -y) \mid x, y, z \in R\} \subseteq R^2$ with the usual addition and scalar multiplication as in R^2 and associated statements given below. Choose the correct options.

☐

V is closed under addition i.e. $(v_1, v_2 \in V \implies v_1 + v_2 \in V)$.

☐

V is closed under scalar multiplication i.e. $(c \in R, v \in V \implies cv \in V)$.

$(c \in R, v \in V \implies cv \in V)$.

☐

V has a zero element with respect to addition. i.e., there exists some element 0_0 such that $v + 0 = v$ for all $v \in V$.

☐

V is not a vector space.

☐

$(a + b)v = av + bv$ where $a, b \in R$ and $v \in V$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

V is closed under addition i.e. $(v_1, v_2 \in V \implies v_1 + v_2 \in V)$.

V is closed under scalar multiplication i.e. $(c \in R, v \in V \implies cv \in V)$.

$(c \in R, v \in V \implies cv \in V)$.

V has a zero element with respect to addition. i.e., there exists some element 0_0 such that $v + 0 = v$ for all $v \in V$.

$(a + b)v = av + bv$ where $a, b \in R$ and $v \in V$.

1 point

Consider a set $V = \{(1, x) \mid x \in R\} \subseteq R^2$ with the usual addition and scalar multiplication as in R^2 and associated statements given below. Choose the correct options.

☐

V is closed under addition.

☐

V has no zero element with respect to addition. i.e., there does not exist any element 0_0 such that $v + 0 = v$ for all $v \in V$.

☐

$(a + b)v = av + bv$ where $a, b \in R$ and $v \in V$.

☐ None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

V has no zero element with respect to addition. i.e., there does not exist any element 0_0 such that $v + 0 = v$ for all $v \in V$.

$(a + b)v = av + bv$ where $a, b \in R$ and $v \in V$.

1 point

Which of the following sets with the given addition and scalar multiplication (scalars are real numbers in every case) form vector spaces?

$$V_1 = \{(x, y, z) \mid x, y, z \in R\}$$

$$\text{Addition: } (x_1, y_1, z_1) + (x_2, y_2, z_2) = (x_1 + x_2, y_1 + 1, z_2 + 1);$$

$$(x_1, y_1, z_1), (x_2, y_2, z_2) \in V_1$$

$$\text{Scalar multiplication: } c(x, y, z) = (x, y, cz); (x, y, z) \in V_1, c \in R$$

$\{(x, y, z) \mid x, y, z \in \mathbb{R}\}$ Addition: $(x_1, y_1, z_1) + (x_2, y_2, z_2) = (x_1 + x_2, y_1 + 1, z_2 + 1);$
 $(x_1, y_1, z_1), (x_2, y_2, z_2) \in V_1$ Scalar multiplication: $c(x, y, z) =$
 $(x, y, cz); (x, y, z) \in V_1, c \in \mathbb{R}$

$V_2 = \{(x, y, z) \mid x, y, z \in \mathbb{R}\}$
 Addition: $(x_1, y_1, z_1) + (x_2, y_2, z_2) = (x_1, y_1, z_1);$
 $(x_1, y_1, z_1), (x_2, y_2, z_2) \in V_2$ $V_2 =$
 Scalar multiplication: $c(x, y, z) = (x, cy, z); (x, y, z) \in V_2, c \in \mathbb{R}$

$\{(x, y, z) \mid x, y, z \in \mathbb{R}\}$ Addition: $(x_1, y_1, z_1) + (x_2, y_2, z_2) = (x_1, y_1, z_1);$
 $(x_1, y_1, z_1), (x_2, y_2, z_2) \in V_2$ Scalar multiplication: $c(x, y, z) =$
 $(x, cy, z); (x, y, z) \in V_2, c \in \mathbb{R}$

$V_3 = \{(x, y, z) \mid x, y, z \in \mathbb{R}\}$
 Addition: $(x_1, y_1, z_1) + (x_2, y_2, z_2) = (x_1 + x_2, 0, 0);$
 $(x_1, y_1, z_1), (x_2, y_2, z_2) \in V_3$ $V_3 =$
 Scalar multiplication: $c(x, y, z) = (cx, y, z); (x, y, z) \in V_3, c \in \mathbb{R}$

$\{(x, y, z) \mid x, y, z \in \mathbb{R}\}$ Addition: $(x_1, y_1, z_1) + (x_2, y_2, z_2) = (x_1 + x_2, 0, 0);$
 $(x_1, y_1, z_1), (x_2, y_2, z_2) \in V_3$ Scalar multiplication: $c(x, y, z) =$
 $(cx, y, z); (x, y, z) \in V_3, c \in \mathbb{R}$

Choose the correct options.

☐

V_3 V_3 is not vector space.

☐

V_2 V_2 is vector spaces.

☐ All are vector spaces.

☐

V_1 V_1 is not vector space.

No, the answer is incorrect.

Score: 0

Accepted Answers:

V_3 V_3 is not vector space.

V_1 V_1 is not vector space.

If (a, b, c) (a, b, c) is the inverse element of $(1, 2, 5) \in \mathbb{R}^3$ $(1, 2, 5) \in \mathbb{R}^3$ of the vector space $\mathbb{R}^3$,
 then find the value of $a - b + c$ $a - b + c$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) -4

1 point

Level 2

1 point

Consider the following sets:

- $V_1 = \{A \mid A \in M_{2 \times 2}(\mathbb{R}) \text{ and } A \text{ is a symmetric matrix, i.e., } A = A^T\}$ $V_1 = \{A \mid A \in M_{2 \times 2}(\mathbb{R}) \text{ and } A \text{ is a symmetric matrix, i.e., } A = A^T\}$

- $V_2 = \{A \mid A \in M_{2 \times 2}(R) \text{ and } A \text{ is a scalar matrix}\}$ $V_2 = \{A \mid A \in M_{2 \times 2}(R) \text{ and } A \text{ is a scalar matrix}\}$
- $V_3 = \{A \mid A \in M_{2 \times 2}(R) \text{ and } A \text{ is a diagonal matrix}\}$ $V_3 = \{A \mid A \in M_{2 \times 2}(R) \text{ and } A \text{ is a diagonal matrix}\}$
- $V_4 = \{A \mid A \in M_{2 \times 2}(R) \text{ and } A \text{ is an upper triangular matrix}\}$ $V_4 = \{A \mid A \in M_{2 \times 2}(R) \text{ and } A \text{ is an upper triangular matrix}\}$
- $V_5 = \{A \mid A \in M_{2 \times 2}(R) \text{ and } A \text{ is a lower triangular matrix}\}$ $V_5 = \{A \mid A \in M_{2 \times 2}(R) \text{ and } A \text{ is a lower triangular matrix}\}$

Choose the set of correct options.

☐

Only V_1 V_1 is a subspace of $M_{2 \times 2}(R)$ $M_{2 \times 2}(R)$.

☐

Only V_4 V_4 is a subspace of $M_{2 \times 2}(R)$ $M_{2 \times 2}(R)$.

☐

Both V_2 and V_3 V_2 and V_3 are subspaces of $M_{2 \times 2}(R)$ $M_{2 \times 2}(R)$

☐

All are subspaces of $M_{2 \times 2}(R)$ $M_{2 \times 2}(R)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

Both V_2 and V_3 V_2 and V_3 are subspaces of $M_{2 \times 2}(R)$ $M_{2 \times 2}(R)$

All are subspaces of $M_{2 \times 2}(R)$ $M_{2 \times 2}(R)$

1 point

Choose the set of correct options.

☐

A vector in R^3 R^3 can be thought of as matrix of order 1×3 or 3×1 1×3 or 3×1 .

☐

Intersection of any two subspaces of a vector space V V is not a subspace.

☐

Intersection of any two subsets of a vector space V V is a subspace.

☐

The empty subset of a vector space V V is a subspace of V V .

☐

$V = \{(x, 0, 0) \mid x \in R\}$ $V = \{(x, 0, 0) \mid x \in R\}$ is a subspace of R^3 R^3 .

No, the answer is incorrect.

Score: 0

Accepted Answers:

A vector in R^3 R^3 can be thought of as matrix of order 1×3 or 3×1 1×3 or 3×1 .

$V = \{(x, 0, 0) \mid x \in R\}$ $V = \{(x, 0, 0) \mid x \in R\}$ is a subspace of R^3 R^3 .

1 point

Consider a set $V = \{(x, y) \mid x, y \in R\} \subseteq R^2$ $V = \{(x, y) \mid x, y \in R\} \subseteq R^2$ with the usual addition as in R^2 R^2 and scalar multiplication is defined as

$$c(x, y) = \begin{cases} (0, 0) & c = 0 \\ (cx, cy) & c \neq 0 \end{cases} \quad (x, y) \in V, \quad c \in R \quad c(x, y) = \begin{cases} (0, 0) & c = 0 \\ (2cx, cy) & c \neq 0 \end{cases} \quad c \in R$$

$(x, y) \in V, \quad c \in R$

Consider the statements given below.

- P: V is closed under addition.

- Q:Q: V has zero element with respect to addition. i.e., there exists some element 0 such that $v + 0 = v$ for all $v \in V$.
- R:R: $1.v = v$ where $1 \in R$ and $v \in V$.
- S:S: $a(v_1 + v_2) = av_1 + av_2$ where $v_1, v_2 \in V$ and $a \in R$.
- T:T: $(a + b)v = av + bv$ where $a, b \in R$ and $v \in V$.

Choose the set of correct options.

- ☐ Only P is true.
☐ Only Q is true.
☐ Both P and Q are true.
☐ P, Q and S are true.
☐ Both R and T are not true.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Both P and Q are true.

P, Q and S are true.

Both R and T are not true.

Consider the following statements

- Statement P: Statement P: $W = \{A \mid A \in M_{2 \times 2}(R) \text{ and } A^2 = A\}$, a subset of $M_{2 \times 2}(R)$ with usual addition and scalar multiplication as in $M_{2 \times 2}(R)$, is a subspace of $M_{2 \times 2}(R)$.
- Statement Q: Statement Q: $W = \{A \mid A \in M_{2 \times 2}(R) \text{ and } \det(A) = 0\}$, a subset of $M_{2 \times 2}(R)$ with usual addition and scalar multiplication as in $M_{2 \times 2}(R)$, is a subspace of $M_{2 \times 2}(R)$.
- Statement R: Statement R: Consider a system of linear equations $Ax = b$, where A is invertible square matrix of order 2, $x = (x_1, x_2)$ and $b \neq 0$. Let $W = \{x = (x_1, x_2) \mid x \in R^2 \text{ and } Ax = b\}$ is a subset of R^2 with usual addition and scalar multiplication as in R^2 , is a subspace of R^2 .
- Statement S: Statement S: Consider a system of linear equations $Ax = 0$, where A is invertible square matrix of order 2 and $x = (x_1, x_2)$. Let $W = \{x = (x_1, x_2) \mid x \in R^2 \text{ and } Ax = 0\}$ is a subset of R^2 with usual addition and scalar multiplication as in R^2 , is a subspace of R^2 .

Find the number of correct statements

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point